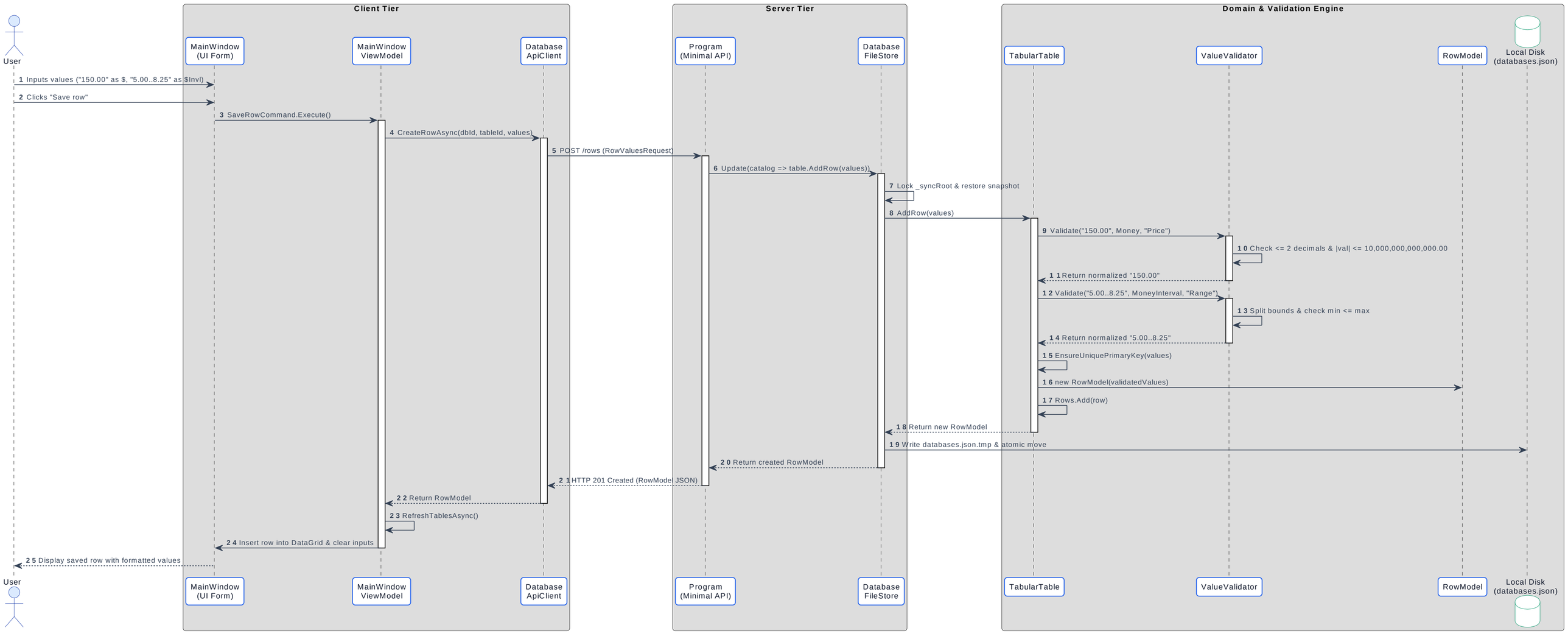
Client Tier
Server Tier
Domain & Validation Engine
User
MainWindow (UI Form)
MainWindow ViewModel
Database ApiClient
Program (Minimal API)
Database FileStore
TabularTable
ValueValidator
RowModel
Local Disk (databases.json)
1 Inputs values ("150.00" as $, "5.00..8.25" as $Invl)
2 Clicks "Save row"
3 SaveRowCommand.Execute()
4 CreateRowAsync(dbId, tableId, values)
5 POST /rows (RowValuesRequest)
6 Update(catalog => table.AddRow(values))
7 Lock _syncRoot & restore snapshot
8 AddRow(values)
9 Validate("150.00", Money, "Price")
10 Check <= 2 decimals & |val| <= 10,000,000,000,000.00
11 Return normalized "150.00"
12 Validate("5.00..8.25", MoneyInterval, "Range")
13 Split bounds & check min <= max
14 Return normalized "5.00..8.25"
15 EnsureUniquePrimaryKey(values)
16 new RowModel(validatedValues)
17 Rows.Add(row)
18 Return new RowModel
19 Write databases.json.tmp & atomic move
20 Return created RowModel
21 HTTP 201 Created (RowModel JSON)
22 Return RowModel
23 RefreshTablesAsync()
24 Insert row into DataGrid & clear inputs
25 Display saved row with formatted values